A First Lean Formalization Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

May 29, 2026

Abstract

This note records the present state of a Lean formalization of a narrow part of Mochizuki's inter-universal Teichmuller theory: the passage from IUT III, Theorem 3.11 to Corollary 3.12, with attention to the Scholze–Stix objection. The current Lean code proves conditional route theorems and obstruction lemmas. It does not prove Corollary 3.12 and does not refute it. The main formal result so far is that a nonarchimedean (Ind3) entry, together with explicit Hodge/SHE square-weight transport data, feeds a hull/log-volume q-to-Θ comparison, while several simplified collapse mechanisms are rejected by Lean.

1 Scope

The review used the local Markdown conversions of IUT I–IV, Mochizuki's 2026 formalization note, and the Scholze–Stix note. The relevant source passages are:

- IUT I: initial Θ-data, prime strips, Hodge theaters, and non-scheme-theoretic Θ-links.
- IUT II: Hodge–Arakelov evaluation, theta values of shape q^{j^2}, multiradiality, and Gaussian monoids.
- IUT III: Theorem 3.11; Corollary 3.12, especially Steps (x) and (xi); Remarks 3.12.2 and 3.12.3.
- IUT IV: later explicit log-volume estimates and Diophantine consequences.
- Scholze–Stix: the ordered real-line identification problem in their Section 2.2.
- Mochizuki 2026: the "3.11.5 $\Rightarrow$ 3.12" fourth-triangle decomposition.

2 Mathematical Target

IUT III Step (x) treats procession-normalized log-volumes as invariants for (Ind1) and (Ind2), and turns (Ind3) into an upper inequality. Step (xi) then uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the q-pilot log-volume in an upper ray controlled by the Θ-pilot log-volume:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|} \quad \Rightarrow \quad -|\log(q)| \leq -|\log(\Theta)|.$$

Mochizuki's 2026 note isolates the final comparison as:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The present formalization targets only this corridor, not the full construction of Theorem 3.11.

3 Dispute Interface

Scholze–Stix require explicit bookkeeping for all ordered one-dimensional real vector spaces. Their pressure point is that the Θ-side concrete objects carry j^2-scaled degrees, while a consistent identification of the real lines appears to leave no room for inserting all these scalars.

The formalization therefore separates:

$$\begin{array}{c} \text{representative } j^2 \text{ data} \\ \text{balanced sign-compatible data} \\ \text{aggregate averaged data} \\ \text{final hull/log-volume data} \end{array} \quad \text{and} \quad \begin{array}{c} \text{raw real equality} \\ \text{transported equality} \\ \text{matched transport scales} \\ \text{cancellable ordered-line comparison.} \end{array}$$

This separation is the main correctness guard in the current code.

4 Lean Coverage

The relevant implementation is in

`Iut/Stage1/IUTStage1Source.lean` and `Iut/Stage1/IUTStage1Experiments.lean`.

The code currently covers:

- labeled real-line copies and positive-scale transports;
- a finite $\mathbb{F}_\ell$-label model with representative square weights $j \mapsto j^2$;
- a representative pilot-degree profile $\deg(\Theta_j) = j^2 \deg(q)$;
- proof that one label-independent scale cannot absorb all representative j^2 scales in this model;
- proof that, for nonzero $\deg(q)$, one label-independent scale cannot account for all representative Θ_j-degrees;
- proof that the raw representative j^2 theta-degree rule does not descend to the sign quotient $|F_\ell| = F_\ell/\{\pm 1\}$ when $\deg(q) \neq 0$;
- an absolute-label exponent on $|F_\ell| = \{0\} \cup F_\ell^\times/\{\pm 1\}$ which agrees with j^2 for the canonical representatives $0 \le j \le (\ell - 1)/2$, and the corresponding theta-degree profile;
- the degree-level Gaussian evaluation $q \mapsto (q^{j^2})_{j \in |F_\ell|}$, including identity at $j = 1$;
- a log-volume shadow of the IUT III splitting-monoid action: the generator attached to a procession label contributes $j^2 \deg(q)$, contributes 0 at the core label, and adds its finite generator sum to the tensor-packet log-volume;
- the finite procession containers $S^\pm_{j+1} = \{0, \ldots, j\}$, their nested inclusions, core-label stability, stage cardinality $j + 1$, and total procession ambiguity $(\ell^\pm)!$ rather than $(\ell^\pm)^{\ell^\pm}$;
- the map from the final procession container to $|F_\ell|$, sending the core label to 0 and recovering the exponent j^2;
- the log-volume shadow of the tensor packet over $S^\pm_{j+1}$, with each factor a finite-fiber direct sum over places above $v_{\mathbb{Q}}$, total log-volume given by the finite sum over the container, normalized denominator $j + 1$, and invariance under container permutations;

- base-valuation products of tensor packets, the finite $(F^{\times}_{\mathrm{mod}})_j$-action shadow, and a coric transfer shadow $F \to D \to D^{\vdash}$ preserving product log-volume without identifying Hodge-theater histories;
- the Step (xi-d)–(xi-g) shadow: determinant normalization, upper-ray membership, and equality of the two q-pilot log-volume computations;
- the Step (xi-h) warning: for $N \geq 2$ and $\Theta < 0$, the tensor-power boundary satisfies $N\Theta < \Theta$;
- the Step (xii) shadow: local Frobenioid p_v^N-ambiguity changes log-volume by an integer shift, while global calibration fixes exponent 0;
- the Remark 3.12.1(iii) arithmetic fact $\mathbb{Q}_{>0} \cap \mathbb{Z}^{\times} = \{1\}$;
- the Remark 3.12.1(ii) generalized-link numerical bound $C_\Theta \geq -\lambda$ for $\lambda \in \mathbb{Q}_{>0}$, with strict strengthening over the $\lambda = 1$ bound when $\lambda < 1$;
- the Remark 3.12.2 distinct-label implication q-itw $\Rightarrow$ q-itw $\wedge$ Θ-itw/indets, with q-native and Θ-native labels proved distinct;
- the Remark 3.12.2 toy calculation: forgotten concrete assignments $-h$ and $-2h$ are incompatible for $h > 0$, while $-h \in \mathbb{R}_{\leq -2h+\epsilon}$ implies $h \leq \epsilon$;
- the Fig. 3.9 column table: both $\bullet_0$ and $\circ$ roles are required in both columns, while q-pilot multiradiality is absent;
- the Remark 3.12.2(v) absorption distinction: zero-column unit indeterminacy is absorbed by a hull upper ray, while one-column logarithmic conjugacy is absorbed as log-volume equality;
- the Remark 3.12.3(i) metric analogy: $\mathrm{vol}(S) > 0$ and $\chi_S = -\mathrm{vol}(S)$ imply $\chi_S < 0$;
- the Remark 3.12.4(iii) factor-p analogy: theta-label normalization divides a gap log-volume by ℓ, and multiplication by ℓ recovers it;
- the Remark 3.12.4(v) p-adic analogue: an inclusion, not isomorphism, gives $(1-p)(2g-2) \leq 0$;
- the IUT IV, Theorem 1.10 algebraic handoff: an explicit expression for C_Θ, together with $C_\Theta \geq -1$ and the remaining elementary estimates, implies the displayed $\frac{1}{6}\log(q)$ bound;
- the final elementary coefficient estimates in IUT IV, Theorem 1.10: $(1 - 12/\ell^2)^{-1} \leq 2$ and $(1 - 12/\ell^2)^{-1}\frac{1}{4}(1 + 36d_{\mathrm{mod}}/\ell) \leq 1 + 80d_{\mathrm{mod}}/\ell$;
- the IUT IV, Theorem 1.10 base-change monotonicity: $\log(d_{F_{\mathrm{tpd}}}) + \log(f_{F_{\mathrm{tpd}}}) \leq \log(d_F) + \log(f_F)$ propagates through the final right-hand side;
- the IUT IV, Theorem 1.10 Step (i) identities: $n^{-1}\sum_{m=1}^{n} m = (n+1)/2$ and $n^{-1}\sum_{m=1}^{n} m^2 = (2n+1)(n+1)/6$;
- the IUT IV, Theorem 1.10 Step (ii) small-prime error: $\log(2^{11}3^3 5^2) \leq 21$, represented by the corresponding linear logarithmic bound;
- the IUT IV, Theorem 1.10 Step (i) GL_2 arithmetic constants for $p = 2, 3, 5$, including the product with the quadratic factor from (E5);

- the IUT IV, Theorem 1.10 Step (iii) algebraic estimate combining the Step (ii) K-bound and the displayed $\log(s_Q)$-bound into $(1+4/\ell)\log(d_K, f_K) + (16/\ell)\log(s_Q) \le (1+36d_{\mathrm{mod}}/\ell)S + 2\log(\ell) + 70$;

- the IUT IV, Theorem 1.10 Step (viii) Proposition 1.6 case split: the two cases $e^*_{\mathrm{mod}}\ell \ge \eta_{\mathrm{prm}}$ and $e^*_{\mathrm{mod}}\ell < \eta_{\mathrm{prm}}$ imply the uniform $4/3(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$ bound;

- the IUT IV, Theorem 1.10 Step (viii) final absorption of $2\log(\ell) + 74$ and the prime-product term into $10(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$;

- the IUT IV, Theorem 1.10 final display: the Corollary 3.12 C_Θ handoff, followed by the tripodal-to-F monotonicity, gives the final F-level $\frac{1}{6}\log(q)$ inequality;

- the IUT IV, Theorem A bounded-discrepancy conclusion: boundedness below of $(1+\epsilon)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) - \mathrm{ht}_{\omega_X(D)}$ implies the corresponding pointwise height inequality on points of bounded degree;

- the IUT IV, Corollary 2.2(i) bounded-discrepancy chain: $\frac{1}{6}\log(q_2) \approx \frac{1}{6}\log(q_\forall) \approx \frac{1}{6}\mathrm{ht}_\infty \approx \mathrm{ht}_{\omega_X(D)}$ composes to a direct bounded discrepancy relation;

- bounded-discrepancy transfer lemmas: pointwise upper and lower bounds move across an $\approx$ relation with the expected additive constant shift;

- the IUT IV, Corollary 2.2(ii) condition (C1) prime-scale window: $(\log(q_\forall))^{1/2} \le \ell \le 10\delta(\log(q_\forall))^{1/2}$. Lean keeps the square-root term explicit, proves $\log(q_\forall) \ge 0$, and derives $1 \le 10\delta$ from the nonempty window;

- the IUT IV, Corollary 2.2(ii) condition (C2) inequality chain: $\frac{1}{6}\log(q) \le \frac{1}{6}\log(q_2) \le \frac{1}{6}\log(q_\forall) \le (1+\epsilon_E)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) + C_K$;

- the IUT IV, Corollary 2.2(ii) definition of ϵ_E: $\epsilon_E = (60\delta)^2 h^{-1/2}\log(2\delta h)$, represented with $h^{1/2}$ and $\log(2\delta h)$ as explicit real parameters; Lean proves positivity and $\epsilon_E \ge 5\delta h^{-1/2}$ from $\delta \ge 2$, $h^{1/2} > 0$, and $\log(2\delta h) \ge 1$;

- the IUT IV, Corollary 2.2(ii) final ϵ_E-absorption: from $0 < \epsilon_E \le 1$, Lean proves

  $$\frac{1+\epsilon_E/5}{1-2\epsilon_E/5} \le 1+\epsilon_E, \qquad (1-2\epsilon_E/5)^{-1}C_K/2 \le C_K,$$

  and hence the displayed C2 bound from the preceding denominator expression;

- the IUT IV, Corollary 2.2(ii) comparison $\log \mathrm{diff}_X(x_E) = \log(d_{F_{\mathrm{tpd}}})$ and $\log(f_{F_{\mathrm{tpd}}}) \le \log \mathrm{cond}_D(x_E) \le \log(f_{F_{\mathrm{tpd}}}) + \log(2\ell)$;

- the IUT IV, Corollary 2.2 to Theorem A passage: bounded-discrepancy equivalence $\frac{1}{6}\log(q_2) \approx \mathrm{ht}_{\omega_X(D)}$, together with the C2 bound for $\frac{1}{6}\log(q_2)$, gives a lower bound for $(1+\epsilon_E)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) - \mathrm{ht}_{\omega_X(D)}$;

- comparison levels: pointwise representative, aggregate representative, balanced sign-compatible, and hull/log-volume;

- endpoint audits for the 3.11.5 $\Rightarrow$ 3.12 comparison chart;

- nonarchimedean and archimedean (Ind3) local orientations;

- ordered real-line alignment with cancellation only after scale equality;
- a canonical unit-scale ordered alignment from a nonarchimedean (Ind3) entry;
- a bridge from factored square/full-label SHE obligations to the Hodge-descent (Ind3) route.

5 Current Lean Results

The current first-pass experiment report evaluates to:

$$\begin{aligned}
&\text{route theorem flags} && = \texttt{true},\\
&\text{collapse-obstruction flags} && = \texttt{true},\\
&\text{Gaussian/procession finite-model flags} && = \texttt{true},\\
&\texttt{balancedLevelRejectedAtFinalRouteTheoremAvailable} && = \texttt{true},\\
&\texttt{disputeSettledByCurrentStage} && = \texttt{false}.
\end{aligned}$$

Here the route flags are ordered real-line, nonarchimedean entry, and factored SHE availability. The obstruction flags are raw-cancellation failure, label-independent j^2 failure, and sign-quotient representative descent failure. The Gaussian/procession flags include finite tensor-packet, monoid action, base-valuation product, coric-transfer, determinant-normalization, upper-ray, q-pilot two-computation, tensor-power warning, and global Frobenioid-calibration shadows, plus positive-rational unit rigidity. The generalized-link flag records the rational parameter inequality $C_\Theta \geq -\lambda$. The distinct-label flag records the logical intertwining display in Remark 3.12.2 and rejects collapse of the q-native and Θ-native labels. The toy-model flag records the resulting upper-ray arithmetic $h \leq \epsilon$. The column-table flag records the Fig. 3.9 theta/q similarity-difference split. The absorption flag records the zero-column upper-bound versus one-column equality distinction. The metric flag records the Gauss–Bonnet sign calculation used as analogy for the upper semi-compatibility inequality. The factor-p flag records the F_ℓ-label averaging denominator. The Frobenius-derivative flag records the degree inequality from inclusion. The IUT IV flag records the downstream Theorem 1.10 use of the Corollary 3.12 lower bound on C_Θ, including the final coefficient estimates and the tripodal-to-F log-degree monotonicity. The Step (i) flag records the two elementary sum identities used in the leading-term computation. The Step (ii) small-prime flag records the additive error bound 21. The GL_2 flag records the finite cardinality constants feeding that bound. The Step (iii) flag records the $\log(s_Q)$ coefficient absorption. The Step (viii) flag records the prime-product case split used before the final upper bound for C_Θ, plus the final coarse error absorption. The final display flag records the composed Theorem 1.10 inequality with F-level log-degree terms. The Theorem A flag records the bounded-discrepancy height/log-different/log-conductor inequality. The Corollary 2.2 flag records composition of the displayed bounded-discrepancy chain. The transfer flag records movement of pointwise bounds across bounded-discrepancy equivalence. The Corollary 2.2(C1) flag records the displayed prime-scale window and its elementary consequences. The Corollary 2.2(C2) flag records the displayed three-step logarithmic inequality chain. The ϵ_E-definition flag records the source formula and lower estimate for ϵ_E. The ϵ_E-absorption flag records the final denominator removal in C2. The log-different/log-conductor flag records the bridge from tripodal field terms to curve functions. The Corollary 2.2-to-Theorem A flag records the formal bounded-discrepancy transfer from C2 to the Theorem A inequality. The theorem-level route is:

$$\begin{array}{ccccc}
\text{nonarch. (Ind3) entry} & \rightarrow & \text{canonical ordered } \mathbb{R}\text{-alignment} & \rightarrow & \text{source/target alignment} \\
 & & \downarrow & & \downarrow \\
\text{factored SHE transport} & \rightarrow & \text{Hodge-descent audit} & \rightarrow & q_{\text{signed}} \leq \Theta_{\text{signed}}.
\end{array}$$

The route remains conditional on the displayed hypotheses.

6 Audit Findings

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The strongest positive result is conditional: if the nonarchimedean (Ind3) entry, ordered real-line alignment, and Hodge/SHE square-weight transport audit are supplied, then Lean checks the final signed q-to-Θ inequality.

The strongest negative result is also limited: in the current $\mathbb{F}_\ell$-label model, Lean rejects a single label-independent scale accounting for all representative j^2 factors, and it rejects balanced sign-compatible evidence as the final hull/log-volume comparison level. Lean also rejects direct descent of arbitrary raw representative j^2 theta-degree data to $|F_\ell|$, while accepting the half-range exponent convention stated in IUT II. Lean also checks that an N-th tensor-power boundary is a strict strengthening for $N \geq 2$ and negative pilot log-volume. It also checks the Step (xii) numerical warning: nonzero local Frobenioid shifts change log-volume, whereas global calibration restores the unshifted value. The positive rational unit rigidity fact in Remark 3.12.1(iii) is formalized directly over $\mathbb{Q}$. For generalized links, Lean records that $\lambda < 1$ makes $-\lambda > -1$. Lean also checks that the Remark 3.12.2 simultaneous-intertwining step depends on distinct labels rather than an unlabeled identification. In the accompanying toy model, Lean proves the final upper bound exactly from the upper-ray membership hypothesis. Lean also records that the q-pilot column lacks the theta-side multiradiality used in Theorem 3.11(iii)(c). For Remark 3.12.2(v), Lean separates hull absorption of unit shifts from log-volume equality after logarithmic-conjugate compatibility. For Remark 3.12.3(i), Lean proves the stated negative Euler-characteristic sign from positivity of metric volume. For Remark 3.12.4(iii), Lean checks the ℓ-denominator normalization identity for the theta-label averaging analogy. For Remark 3.12.4(v), Lean proves the displayed nonpositive degree defect from $p \geq 2$ and $g \geq 2$. For IUT IV, Theorem 1.10, Lean proves the algebraic implication from the computed C_Θ expression and $C_\Theta \geq -1$ to the final $\frac{1}{6}\log(q)$ estimate, conditional on the separate elementary estimates. Lean now also proves the two displayed final elementary coefficient estimates from $\ell \geq 5$. It also proves the final monotonic replacement of the tripodal log-degree sum by the larger F-level log-degree sum, conditional on the two component monotonicity inequalities cited from Proposition 1.3(i). Lean also proves the Step (i) sum and square-sum identities over $\mathbb{R}$. For Step (ii), Lean proves the small-prime additive error bound from the displayed logarithmic inequalities. Lean also checks the Step (i) GL_2 constants for $p = 2, 3, 5$ and their product with the quadratic extension factor. For Step (iii), Lean proves the displayed coefficient absorption from the source hypotheses $d_{\mathrm{mod}} \geq 1$, $\ell \geq 5$, $2\log(\ell) \leq \ell$, and nonnegativity of the tripodal log-degree sum. For Step (viii), Lean checks that the two Proposition 1.6 cases imply the uniform prime-product bound used in the final C_Θ estimate. Lean also checks the subsequent absorption into the coefficient 10 of $e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}}$. Combining the Corollary 3.12 handoff with the tripodal-to-F monotonicity, Lean proves the final displayed F-level inequality of Theorem 1.10 at this real-valued shadow level. For Theorem A, Lean records the exact bounded-discrepancy conclusion and proves the equivalent pointwise height bound from the lower-bound hypothesis. For Corollary 2.2(i), Lean defines equality up to bounded discrepancy, proves reflexivity, symmetry, transitivity, nonnegative scaling, and composes the displayed chain from $\log(q_2)$ to $\mathrm{ht}_{\omega_X(D)}$. Lean also proves that upper and lower pointwise bounds transfer across such a bounded-discrepancy equivalence with the appropriate constant shift. For Corollary 2.2(ii)(C1), Lean records the displayed window for ℓ and derives nonnegativity of $\log(q_\forall)$, positivity of $(\log(q_\forall))^{1/2}$, and $1 \leq 10\delta$. For Corollary 2.2(ii), Lean proves the final $\frac{1}{6}\log(q)$ bound from the displayed $\log(q)$, $\log(q_2)$, and $\log(q_\forall)$ chain. Lean records the proof's definition of ϵ_E and verifies the stated lower estimate $\epsilon_E \geq 5\delta h^{-1/2}$. Lean also proves the final ϵ_E-absorption step used just before C2: the intermediate expression with denominator $1 - 2\epsilon_E/5$ is bounded by $(1 + \epsilon_E)$ times the log-degree sum plus C_K. It also proves the two sum inequalities relating the tripodal different and conductor terms to $\log \mathrm{diff}_X + \log \mathrm{cond}_D$. Finally, Lean combines the Corollary 2.2(i) bounded-discrepancy height comparison with the C2 bound to construct the Theorem A lower-bound shadow.

7 Missing Mathematics

The present code does not yet formalize:

- initial Θ-data as in IUT I;
- actual Hodge theaters, Frobenioids, prime strips, log-shells, or Θ-links;
- the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;
- the construction of the actual theta values, Gaussian/splitting monoids, $F^{\times}_{\mathrm{mod}}$-actions, Kummer isomorphisms, mono-analyticization, and Frobenioids behind the present shadows;
- the actual construction of log-shells and tensor packets attached to mono-analytic prime-strips in IUT III, Theorem 3.11;
- the genuine holomorphic hull and determinant operations of Step (xi), beyond the present finite determinant/upper-ray/two-computation/tensor-power shadows;
- the global realified Frobenioids of Step (xii), beyond the present integer log-volume shift model;
- the arithmetic-divisor and local log-volume construction behind the IUT IV Theorem 1.10 numerical estimates, and the Diophantine applications;
- the construction of the normalized height, log-different, and log-conductor functions used in IUT IV, Theorem A;
- the construction of the bounded-discrepancy equivalences asserted in IUT IV, Corollary 2.2(i);
- a proof that the current Hodge/SHE transport obligations follow from Mochizuki's full hypotheses.

8 Next Work

The next significant formal step is not another wrapper. It is to replace current explicit Hodge/SHE transport assumptions by constructions closer to the papers:

$$\begin{gathered}\text{theta values } q^{j^2}\\ \rightsquigarrow\ \text{Gaussian and splitting monoids}\\ \rightsquigarrow\ \text{square/full-label preservation}\\ \rightsquigarrow\ \text{hull/log-volume route.}\end{gathered}$$

Only after this bridge is formalized can the first-pass experiment begin to test the actual mathematical load-bearing part of Mochizuki's response to Scholze–Stix.

References

[1] S. Mochizuki, *Inter-universal Teichmuller Theory I: Construction of Hodge Theaters.*

[2] S. Mochizuki, *Inter-universal Teichmuller Theory II: Hodge-Arakelov-theoretic Evaluation.*

[3] S. Mochizuki, *Inter-universal Teichmuller Theory III: Canonical Splittings of the Log-theta-lattice.*

[4] S. Mochizuki, *Inter-universal Teichmuller Theory IV: Log-volume Computations and Set-theoretic Foundations.*

[5] S. Mochizuki, *On the Formalization of IUT: Preliminary Progress Report*, 2026.

[6] P. Scholze and J. Stix, *Why abc is still a conjecture.*